

# Unlocking History with **ModiTrans**

A Technical Deep-Dive into the 12B-Gemma-Teacher VLM

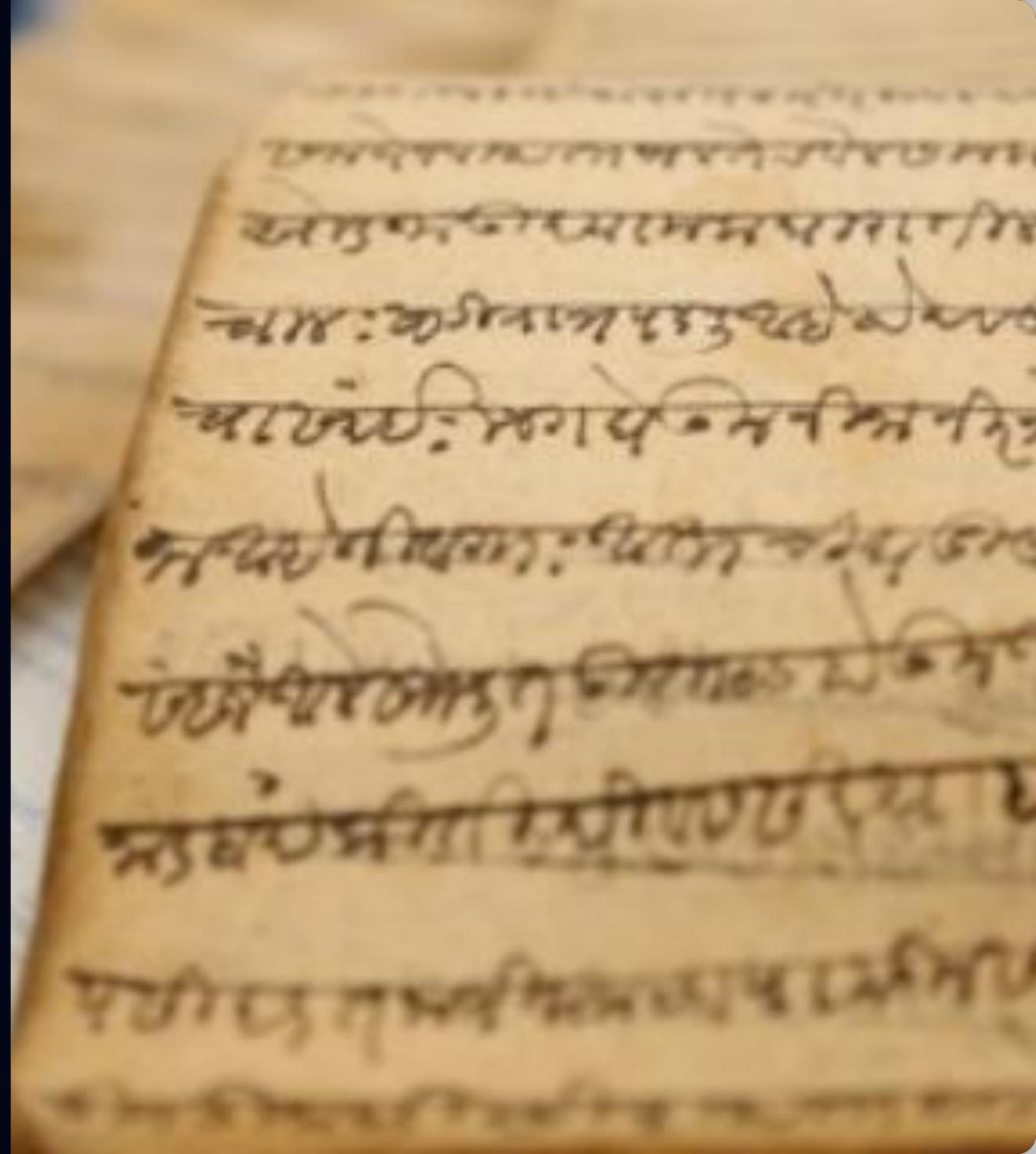
MODEL: historyHulk/ModiTrans-12B-Gemma-Teacher  
TASK: Historic Modi Script to Devanagari Transliteration



# The **Why**: Preservation Crisis

In medieval India, the Marathi language was recorded using the **Modi script**. Today, we face a massive bottleneck:

- ✓ **40 Million** documents remain untransliterated.
- ✓ Extreme shortage of domain experts.
- ✓ Fragile physical condition of records.
- ✓ Critical historical, medical, and land data at risk.





# | The Complexity of Modi Script

Unlike modern Devanagari, Modi script is cursive and features:

- ✓ Continuous pen strokes with few lifts.
- ✓ Significant variation in handwriting styles across eras.
- ✓ Context-dependent ligatures and symbols.
- ✓ No standard "spacing" between words in older manuscripts.

**The Goal:** End-to-end transliteration from Scanned Image directly to Devanagari Text.

| Modi | Gujarati | Devanagari |      |
|------|----------|------------|------|
| ઈ    | ઈ        | ई          | e    |
| ગ    | ગ        | ग          | ga   |
| ઘ    | ઘ        | घ          | gha  |
| ડ    | ડ        | ङ          | da   |
| છ    | છ        | छ          | cha  |
| જ    | જ        | ज          | ja   |
| ઝ    | ઝ        | झ          | nya  |
| ઢ    | ઢ        | ढ          | dha  |
| ત    | ત        | त          | ta   |
| શ    | શ        | श          | sha  |
| ષ    | ષ        | ष          | ssa  |
| ક્ષ  | ક્ષ      | क्ष        | shya |



# | The **Model Tree** & Lineage



## **Base Model**

**Google Gemma-3-12B-it.** A state-of-the-art instruction-tuned VLM with 12 billion parameters.



## **Teacher Model**

**ModiTrans-12B.** The high-capacity model that serves as the gold standard for transliteration accuracy.



## **Student Model**

**MoScNet.** A 163x smaller model trained via Knowledge Distillation to achieve mobile-ready speeds.



# | Why Gemma-3-12B?



Gemma

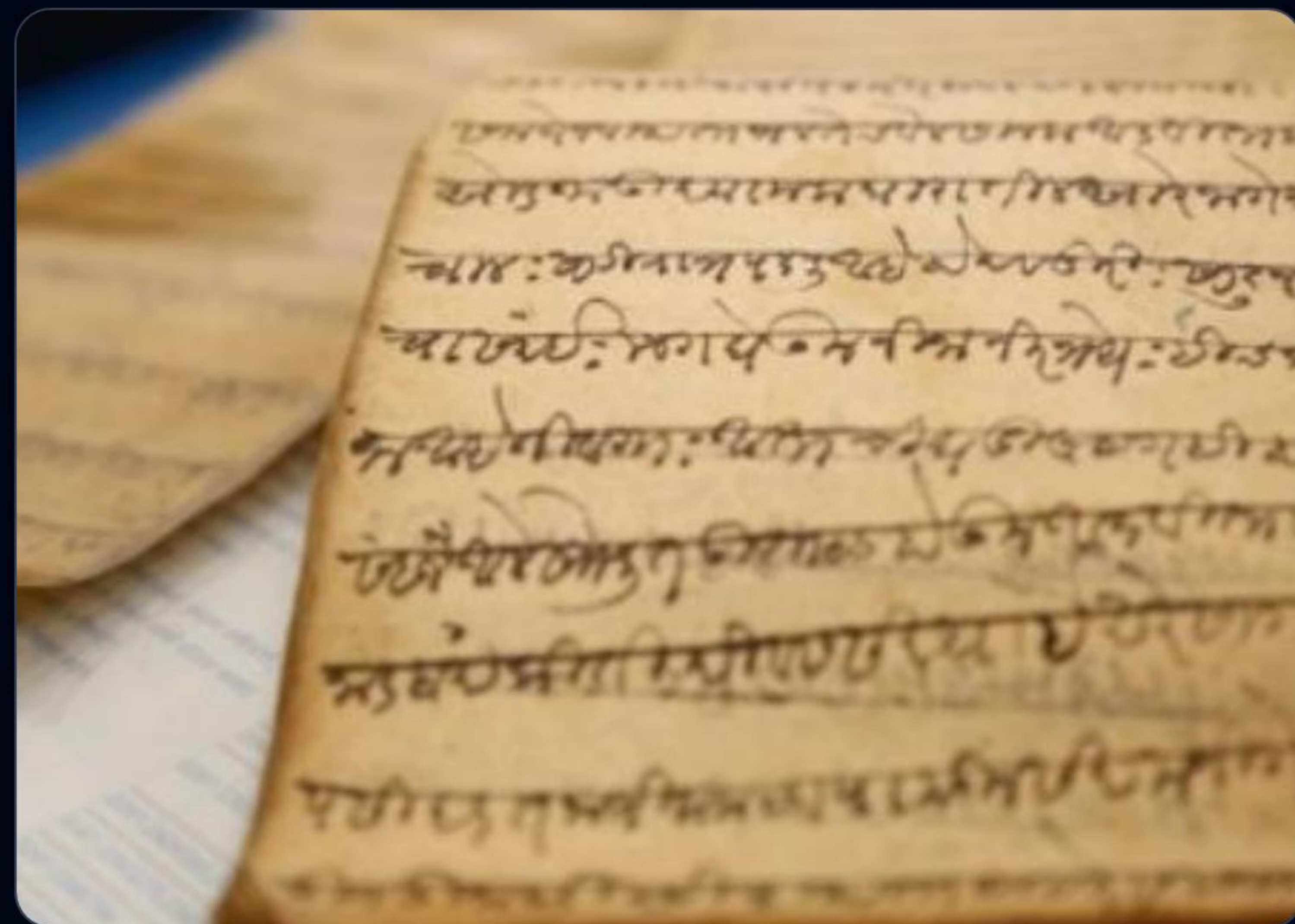
- ✓ **Unified Architecture:** No separate visual encoder; image patches are projected directly into the LLM space.
- ✓ **Parameter Efficiency:** 12B is the "sweet spot" for performance on consumer GPUs.
- ✓ **Large Context:** Supports 128K tokens, vital for long manuscript pages.
- ✓ **Multimodal Native:** Designed from scratch to understand vision and text in a single decoder pass.



# | Dataset 1: MoDeTrans (Real)

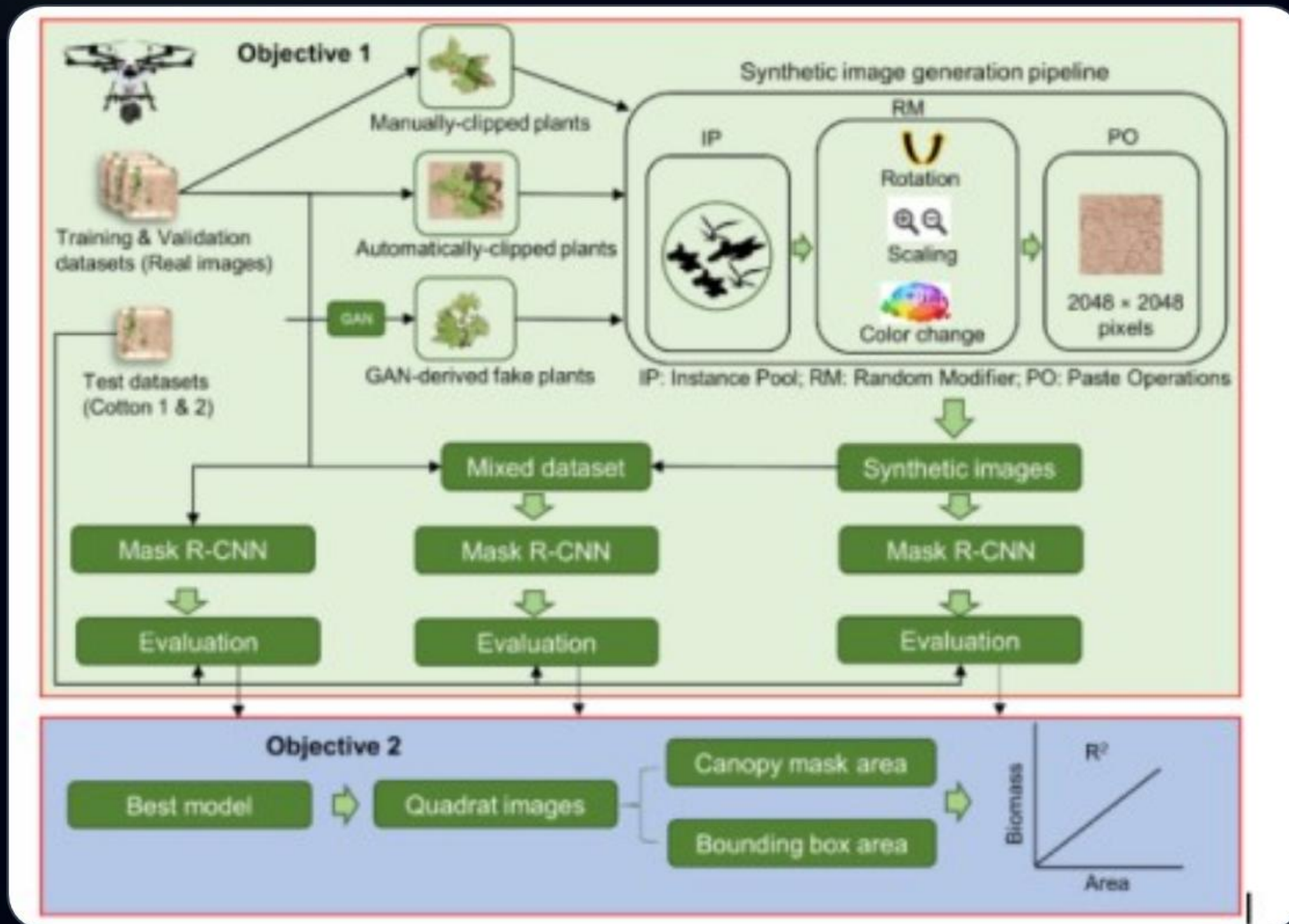
## Authentic Historical Data

- ✓ 2,043 pairs of Modi script images and Devanagari text.
- ✓ Sourced from real historical documents.
- ✓ Includes noise, fading, and varied ink quality.
- ✓ **Preparation:** Manually curated and transliterated by rare domain experts.





# Dataset 2: **SynthMoDe** (Synthetic)



## Programmatic Augmentation

Since 2,000 images aren't enough for 12B parameters, the team generated **Synthetic Data**.

### Synthesis Methodology:

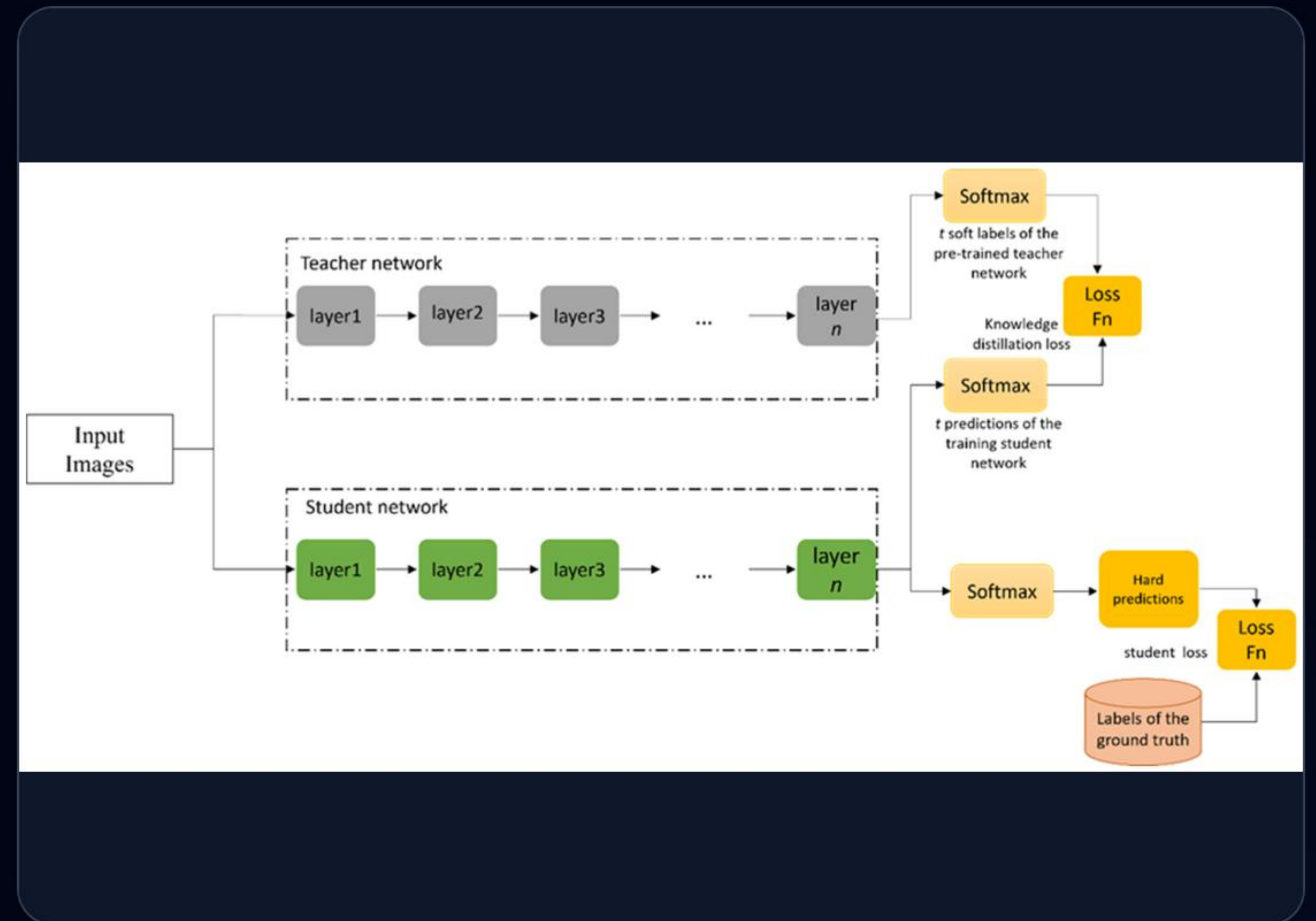
1. Collect Devanagari text (Marathi).
2. Use Modi fonts + Random Distortion.
3. Apply textures (paper, ink bleed).
4. Generate Infinite Training Pairs.



# | The Teacher-Student Strategy

To deploy on edge devices, the I2B model acts as a "Teacher".

- ✓ **Teacher (I2B):** High accuracy, slow inference.
- ✓ **Student:** Ultra-lightweight, 163x fewer params.
- ✓ **Loss Function:** The student minimizes the difference between its predictions and the Teacher's "soft labels."





# | Training Decisions & PEFT

| Decision   | Implementation             | Reasoning                                                    |
|------------|----------------------------|--------------------------------------------------------------|
| Strategy   | LoRA (Low-Rank Adaptation) | Reduce VRAM footprint while keeping 12B base frozen.         |
| Precision  | BFloat16                   | Standard for Gemma to maintain numerical stability.          |
| Input Size | 1024 x 512                 | Capture fine details of cursive Modi strokes.                |
| Optimizer  | AdamW with Warmup          | Prevent catastrophic forgetting of the base language skills. |



# | Expert **Assessment**: SWOT

## **Strengths**

First direct end-to-end VLM for Modi. Exceptional handling of cursive ligatures.

## **Weaknesses**

Limited diversity in 'Real' dataset (2k images). Vulnerable to heavy physical damage noise.

## **Innovation**

The use of Knowledge Distillation to bring historic transliteration to mobile devices.



# | The **Expert Path** to Replicate

If starting this project from scratch, follow this pipeline:

1. **Phase 1:** Build a high-fidelity synthetic engine (more noise, more fonts).
2. **Phase 2:** Pre-train the VLM on generic OCR tasks before specific Modi tuning.
3. **Phase 3:** Use **RLHF** (Human Feedback) to refine transliteration nuances that synthetic data misses.
4. **Phase 4:** Implement sliding window attention for long scrolls.



# | Questions for the Reviewer

- ✓ **Data Leakage:** Are there any overlapping samples between SynthMoDe and MoDeTrans?
- ✓ **Zero-Shot:** How does the model perform on scripts it wasn't trained on (e.g., Brahmi)?
- ✓ **Expert Bias:** How were the manual ground-truth labels validated for historical accuracy?
- ✓ **Robustness:** What is the Character Error Rate (CER) on water-damaged or burned manuscripts?
- ✓ **Privacy:** Does the model unintentionally memorize sensitive land record data?



# | How to Use: Installation

```
# Install necessary libraries
pip install transformers accelerate peft pillow torch

# Log in to Hugging Face (Gated Access required)
huggingface-cli login
```

**Note:** This model is **publicly available** but **Gated**. You must request access on the [historyHulk Hugging Face](#) page.



# | How to Use: Inference

```
from transformers import AutoProcessor, AutoModelForImageTextToText
from peft import PeftModel
import torch

model_id = "google/gemma-3-12b-it"
peft_id = "historyHulk/ModiTrans-12B-Gemma-Teacher"

processor = AutoProcessor.from_pretrained(model_id)
base_model = AutoModelForImageTextToText.from_pretrained(model_id)
model = PeftModel.from_pretrained(base_model, peft_id).to("cuda")

# Process image and prompt
# inputs = processor(images=image, text="Transcribe Modi script...", return_tensors="pt")
# outputs = model.generate(**inputs)
```



# Towards Digital Heritage

ModiTrans is more than a model; it's a bridge to India's medieval scientific and social legacy.

 **Next Step:** Explore the MoScNet Student Model for mobile apps.



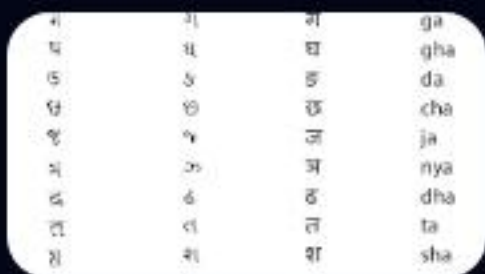
# Image Sources



<https://s.aolcdn.com/hss/storage/midas/461219852cb0efbcf83ea2d7aff8c99c/204763241/2.JPG>

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Source: [shivajiscript.blogspot.com](https://shivajiscript.blogspot.com)

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<https://storage.googleapis.com/gweb-developer-goog-blog-assets/images/Gemma-banner.original.png>

Source: [developers.googleblog.com](https://developers.googleblog.com)

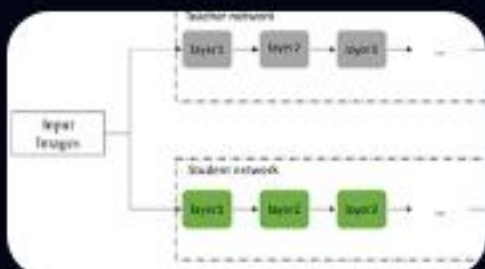
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